

UNIVERSITÀ DEGLI STUDI DI
MILANO - BICOCCA

Numerical Simulations of the Ising model through Monte Carlo Markov Chains



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Abstract

This work presents the numerical study of the ferromagnetic Ising model in one, two, and three dimensions to observe phase transitions as a function of the lattice size. The system configurations were obtained by sampling a Markov Chain at fixed temperature generated with the Metropolis-Hastings algorithm following an analysis of its thermalisation and autocorrelation time. The critical temperature was estimated in the two and three dimensional cases by analyzing the finite size scaling of the magnetization. Analogously, the absence of a finite temperature phase transition was found in the one dimensional case, in accordance with analytical results.

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A brief history of the Ising model

Does this chapter really need to exist?

Chapter 1

Theoretical background

1.1 General overview

The Ising model is a mathematical model of a ferromagnet. [TODO: add context].

The fundamental assumption of the model is that a solid can be represented as a ferromagnet as an array of fixed ions that only interacting with its closest neighbours (and, optionally, an external magnetic field). Such system is described by the classical Hamiltonian

$$H = - \sum_{\langle i,j \rangle} J_{ij} \sigma_i \sigma_j - \mu \sum_i h_i \sigma_i \quad (1.1)$$

where:

- σ_i is the spin value of the ion of the i-th lattice site.
- J_{ij} is a coupling factor between the i-th and j-th lattice sites.
- The sum over $\langle i, j \rangle$ is meant to be over closest neighbours.
- h_i is the magnetic field in the i-th site.
- μ is the magnetic moment of the ions.

This is the special case of the more general quantum Hamiltonian

$$\hat{H} = - \sum_{\langle i,j \rangle} J_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z - \mu \sum_i \left(h_{i,x} \hat{\sigma}_i^x + h_{i,y} \hat{\sigma}_i^y + h_{i,z} \hat{\sigma}_i^z \right) \quad (1.2)$$

where σ_i are replaced by the Pauli matrices of the i-th site.

In this text only $h_{i,z}$ will ever be assumed $\neq 0$, and since if both $h_{i,y}$ and $h_{i,x}$ are null then every operator commute with each other in the canonical basis, which means equations 1.1 and 1.2 are the same.

Given that the quantities $N = \text{number of particles}$, $V = Na = \text{volume of the lattice}$ (where a is the unimportant lattice constant that can be set to 1) are fixed, if the system is in thermal equilibrium with a reservoir at temperature T then the set of microstates of the system is described by the canonical ensemble. In this case, the probability of the system to be in a microstate σ is given by the Boltzmann distribution

$$P_\beta(\sigma) = \frac{e^{-\beta H(\sigma)}}{Z_\beta} \quad (1.3)$$

where Z_β is the canonical partition function

$$Z = \sum_i e^{-\beta E_i} = \sum_{\sigma_i} e^{-\beta \sum_{\langle i,j \rangle} J_{ij} \sigma_i \sigma_j - \beta \mu \sum_i h_i \sigma_i} \quad (1.4)$$

All the same, given an operator $\hat{O}$ its mean is intended to be

$$\langle \hat{O} \rangle_\beta = \sum_\sigma \hat{O}(\sigma) P_\beta(\sigma) \quad (1.5)$$

which is the canonical average.

The minus in front of the first sum in equations 1.1 and 1.2 are conventional. Varying the sign of the coefficients J_{ij} two differently behaving systems are described:

1. $J_{ij} > 0$: ferromagnetic
2. $J_{ij} < 0$: antiferromagnetic
3. $J_{ij} = 0$: non-interacting

Historically, the Ising model is conventionally intended to be ferromagnetic. For simplicity sake, J_{ij} along the constants k_B , $\hbar$, h_i , μ will be set $= 1 \ \forall i, j$, when not null.

1.2 Phase transitions

It's been stated that the system described by the Ising model is represented by the canonical ensemble, which is governed by the Helmholtz free energy. This, like every other thermodynamic potential, can be expressed as a function of the partition function $Z = Z(T, h)$:

$$F = E - TS = -\frac{\ln(Z)}{N\beta} \quad (1.6)$$

Formally, a phase transition is defined as a non-analyticity in the free energy in the thermodynamic limit. To observe such a non-analyticity one can study the partial derivative

$$M = -\frac{\partial F}{\partial H} \quad (1.7)$$

that takes the name of magnetization.

This is a handy observable in the numerical approach because its average is easily obtainable by averaging the spin values of the lattice sites:

$$\langle M \rangle = \frac{1}{N} \sum_i \sigma_i \quad (1.8)$$

and studying its behaviour as a function of the temperature T and the external magnetic field h one can identify the phase transitions.

It should be noted that this definitions formally hold even in the case of no external magnetic field. This is true only in the thermodynamic limit, where the system is infinitely large and the magnetization can be non-zero even when h vanishes.

In a finite system the Hamiltonian is symmetric under the transformation $\sigma_i \rightarrow -\sigma_i$, and this causes the magnetization to be zero when $h = 0$ because

$$\langle M \rangle = \left. \frac{\partial \frac{\ln(Z)}{\beta N}}{\partial h} \right|_{h=0} = 0 \quad \forall T$$

The solution to this problem is

[TODO: i fear this can be greatly expanded on]

1.3 Solutions

1.3.1 One dimensional case

The one dimensional Ising model is exactly solvable with and without an external magnetic field. The magnetization is given by:

[TODO: discuss solutions]

1.3.2 Two dimensional case

1.3.3 Three dimensional case

Chapter 2

Methods

2.1 Data representation

The code used in this project was written in python, using the Numba just-in-time compiler to achieve optimal execution speed (along best practices and algorithm optimizations in writing the code). From a development standpoint, every single Ising model configuration is represented in python as a d dimensional numpy array of shape N repeated d times.

```
1
2 def new_random_ising(N: tuple[int, ...]):
3     return np.ones(N, dtype = np.int8)
4
5 m = new_random_ising(tuple([N] * dim))
6
```

Note that the data type that the numpy array contains is `np.int8`. In order to minimize storage space, the idea of using boolean variables to represent 1 and -1 was considered, but was ultimately ditched in favour of storing integers. The latter grants faster executions when calculating physical observables (e.g.: energy, magnetization) at the loss of some disk space.

Every one of these numpy array is generated and stored in another numpy array of arbitrary length that represents the Markov Chain. This is done for a number of temperatures around the expected critical temperature, and is repeated for every lattice size under scrutiny and for every dimension. Ultimately, this data is stored in a hdf5 file for efficient i/o operations and in order to contextually add metadata and processed data (e.g.: autocorrelation time, thermalisation time, etc.) to the raw chain.

For every combination of dimension and lattice size, the generated row data is stored in a numpy array of shape $(N_{temps}, N_{steps}, [N_{dim}] \times d)$.

[TODO: maybe discuss floating point precision]

2.2 Monte Carlo Markov Chains

2.2.1 Generation

To study a system at a given temperature it is necessary to be able to generate a random configuration of said system. These random configurations are distributed as a function of their energy: as discussed the probability of a system of being in a state with energy E in the canonical ensemble is given by:

$$P(E) = \frac{\Omega(E)e^{-\beta E}}{Z} \quad (2.1)$$

Now, it would be ideal to write this probability analytically: to do that we would need to know how to write both $\Omega(E)$ and Z , but such an operation is, unfortunately, simply not possible. The lone computing of Z would require summing over all possible configurations of the system, which in turns would mean to compute the energy of every single configuration. The number of possible configurations of a (Nd) spins lattice is $2^{(Nd)}$ and computing the energy of a single configuration is an $O(N)$ operation, which means that the overall complexity of computing Z is $\mathcal{O}(N2^{(Nd)})$, which is clearly not viable for any reasonable value of N .

The solution to this problem is a Markov Chain Monte Carlo method, that is a method for generating a sequence of independent samples from a desired distribution without knowing the normalization constant. One algorithm that produces such a chain is the Metropolis-Hastings algorithm. This algorithm can draw samples from any probability distribution with probability density $P(x)$, provided that we know another function proportional to the density P that we're able to compute analitically (in this case $e^{-\beta E}$) so that we can write $P(E) \propto e^{-\beta E}$. The target distribution P is also called posterior.

The algorithm's idea is to start from a random configuration of the system and then to move to another configuration of the system with a set probability which is determined by the energy distribution.

The formal steps of the Metropolis algorithm in this use case are:

1. Start from a random configuration of the system x .
2. Randomly select a spin (or a number of spins) and flip it. We call this new configuration x' .
3. Compute the energy difference of the two systems ΔE .
4. Set an acceptance probability parameter α based on the energy change and the temperature: $\alpha = \frac{T(x' \rightarrow x)P(x')}{T(x \rightarrow x')P(x)}$ where P is the Boltzmann distribution. This is the

core of the algorithm and it's what allows the Markov Chain to converge to the desired distribution.

5. Accept the new configuration with probability $\min(1, \alpha)$, otherwise reject it and keep the current configuration. That is, if the new configuration is more probable than the current one, accept it, otherwise accept it with probability α (that is, a probability proportional to the ratio of the two probabilities).
6. Add the current configuration to the Markov Chain.
7. Repeat steps 2-6 for a number of iterations. The elements of the chain after enough iterations are distributed as $e^{-\beta E}$

The "trick" that makes this algorithm work is that computing α both Z and the Ω terms cancel out.

[TODO: maybe discuss (formally or not) why the sample set given by the MC converges]

Note that randomly selecting a spin and flipping it (step 2) is effectively moving to another random configuration of the system close to the current one. This movement is interpreted as a random sample from a distribution centered around the current configuration. Picking a random spin to flip is an arbitrary choice, and other methods could be used to select a neighboring configuration, but has the advantage that the probability of moving from x to x' ($T(x \rightarrow x')$) is the same as moving from x' to x ($T(x' \rightarrow x)$), which is the Hastings condition of the algorithm. This condition is not strictly necessary, but it simplifies the computation of $\alpha = \frac{e^{-\beta E'}}{e^{-\beta E}} = e^{-\beta \Delta E}$ to be only dependent on the energy change and the temperature.

```

1
2 def metropolis_hastings(model: np.ndarray, T: float, steps: int):
3     '''
4         Generate a Markov Chain of configurations of the Ising model
5         ↪ at a given temperature T.
6     '''
7     curr_model = np.copy(model)
8     models_samples = np.empty(
9         (steps,) + curr_model.shape,
10        dtype = curr_model.dtype
11    )
12
13    N = curr_model.shape[0]
14
15    for i in range(steps):
16        idx0 = np.random.randint(0, N)
17        spin = curr_model[idx0]
18        deltaE = compute_energy_change(curr_model, idx0)
19

```

```

20     accept_probability = np.minimum(1.0,
    ↪     target_distribution(deltaE, T))
21     if np.random.random() <= accept_probability:
22         curr_model[idx0] = -curr_model[idx0]
23
24     models_samples[i] = curr_model
25
26     return models_samples
27

```

Also note that since α depends only on the relative energy change the computation can be greatly optimized by computing only the energy change given by the single spin flip, instead of computing the energy of two entire configurations.

In figure 2.1 some of the first steps of the chain can be seen as starting from different configurations (every spin set at random, all spins up, all spins down). Note that all of the chains start far from what should be the center of the distribution. This assures that before a single sample can be considered randomly sampled from the posterior some "adjusting" time needs to pass. More generally, such a chain suffers from some kind of "recency bias", meaning that, even though the exact previous step at any point of the chain is not ever identifiable, the approximate location (in energy) in of the configuration needs some time to be allowed to properly change. This is shown in the chains by the macro clusters of samples that remains in a narrow vertical space.

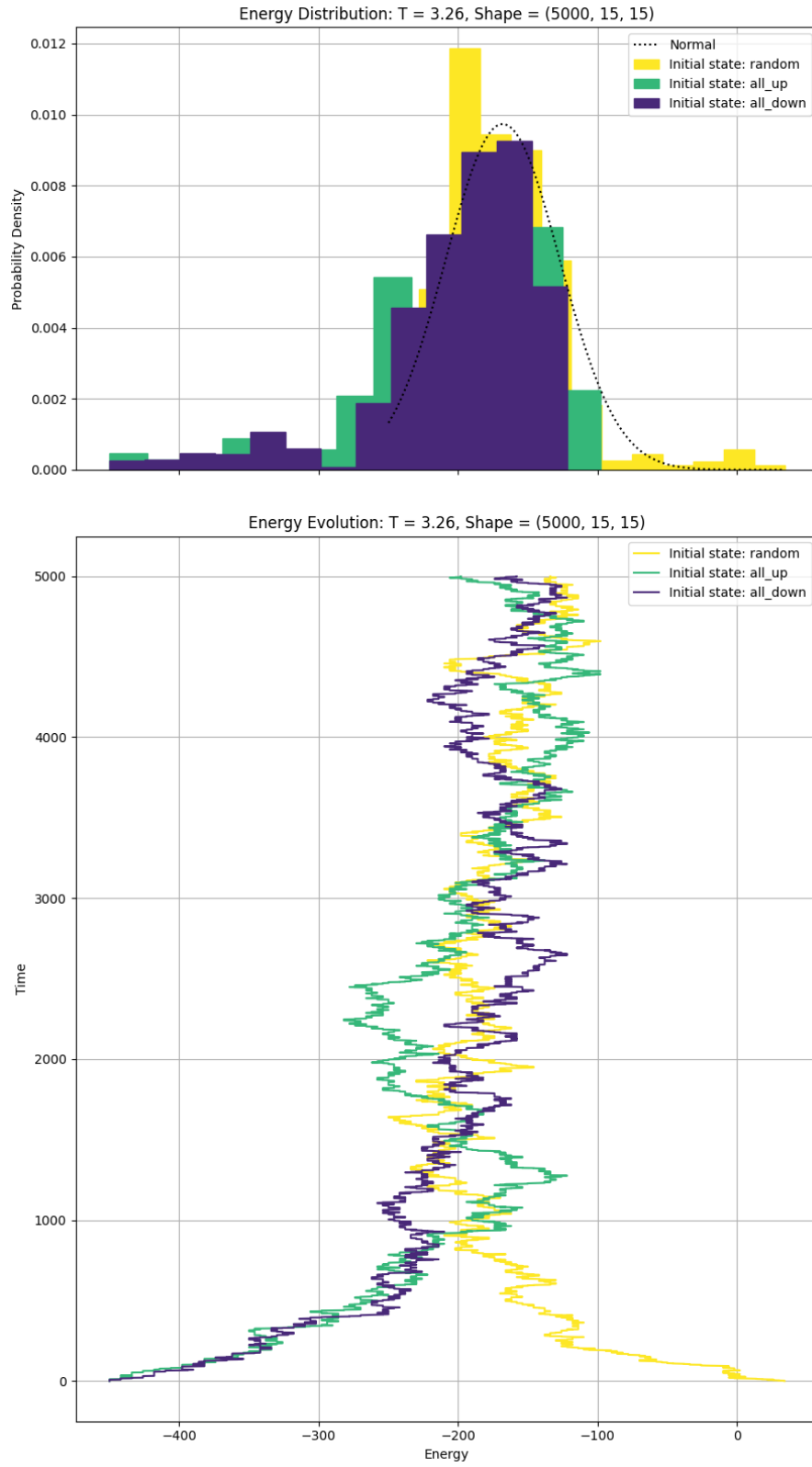


Figure 2.1: Markov Chain and relative distribution of the energy of the system.

Figure 2.2 shows the very same chains after a much larger number of steps. It's clear that the chains have converged to a stationary distribution, and that the initial configuration is no longer relevant.

The energy distribution of the elements in the chain can now be appreciated to be converging to a normal distribution. This is not in contradiction with the fact that the distribution is supposed to be a Boltzmann distribution, but is in fact a consequence of the Central Limit Theorem (CLT), which for temperatures far from the critical temperature is applicable. On the contrary, close to T_C CLT is not applicable and the distribution diverges. [TODO: this can be expanded on]

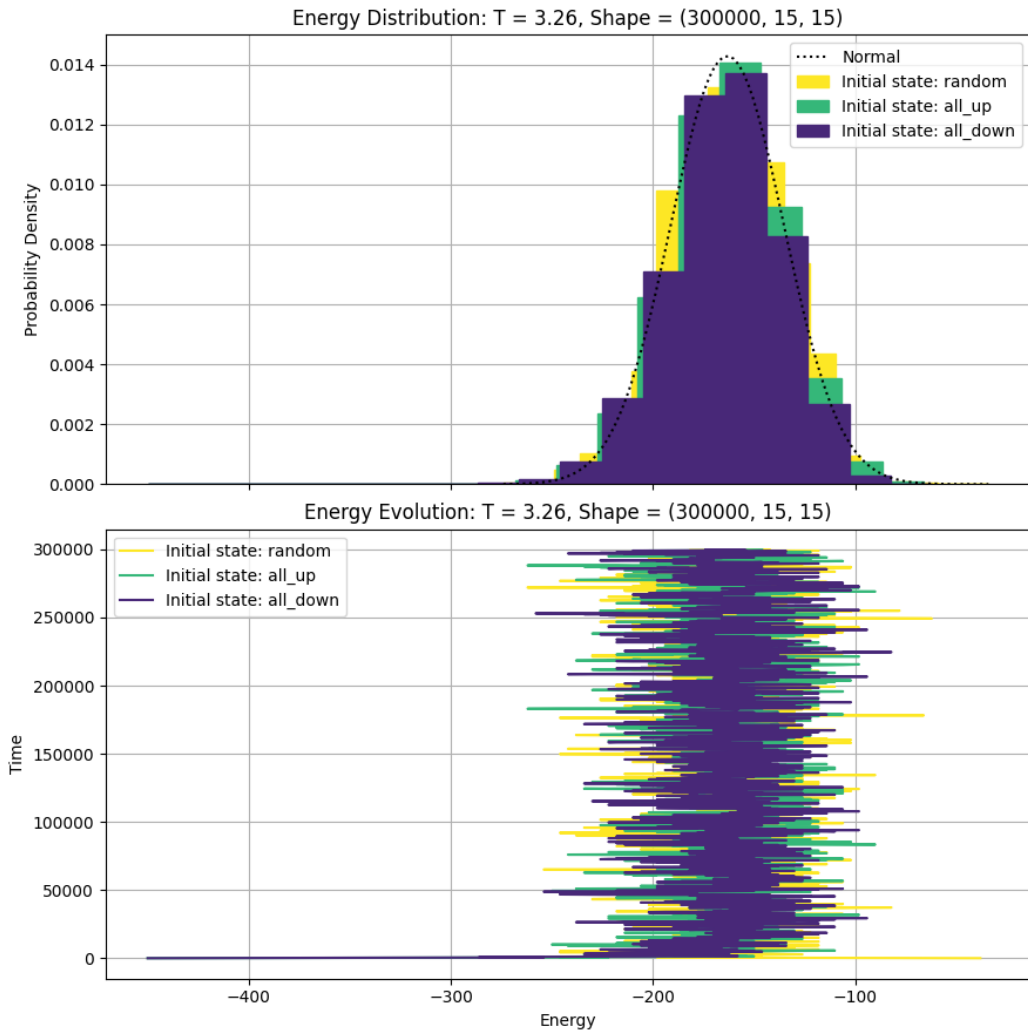


Figure 2.2: Markov Chain and relative distribution of the energy of the system.

2.2.2 Analysis

As stated, a Markov Chain suffers of a recency bias, and the set of configurations of the chain is not very useful as a real sample set since the elements are not independent: on the contrary, they are very closely correlated and the presence of correlations between samples would lead to biased results during analysis.

The extraction of decorrelated data is a problem composed of two parts: the thermalisation and the autocorrelation time. The first one is the number of steps that the Markov Chain needs to reach a stationary distribution, while the second one is the number of steps that need to exist between successive samples before they can be considered independent.

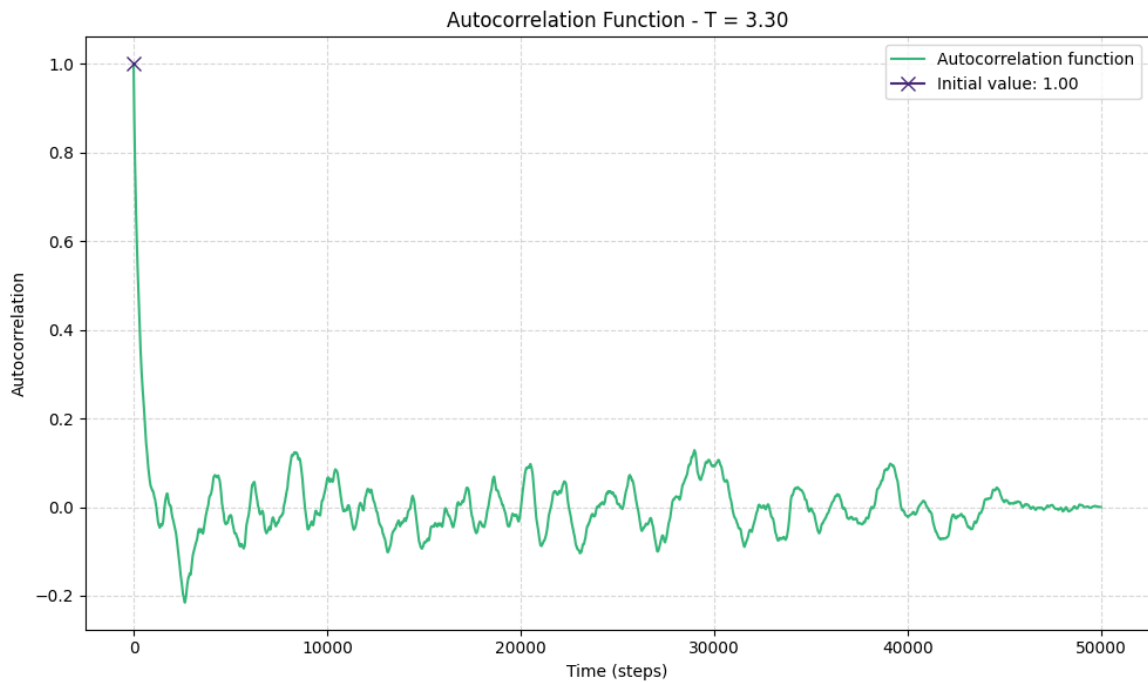


Figure 2.3: Example of autocorrelation function for a Markov Chain.

[TODO: discussion of the autocorrelation function and its properties. An exponential fit after a certainb number of steps is only fitting noise.]

NOTA: Figure 2.3 prove the need of selecting a proper range to fit over. This is given by the linear portion of the following plot.

[TODO: Didnt manage to make use of this]

[TODO: discussion of Sokal's method]

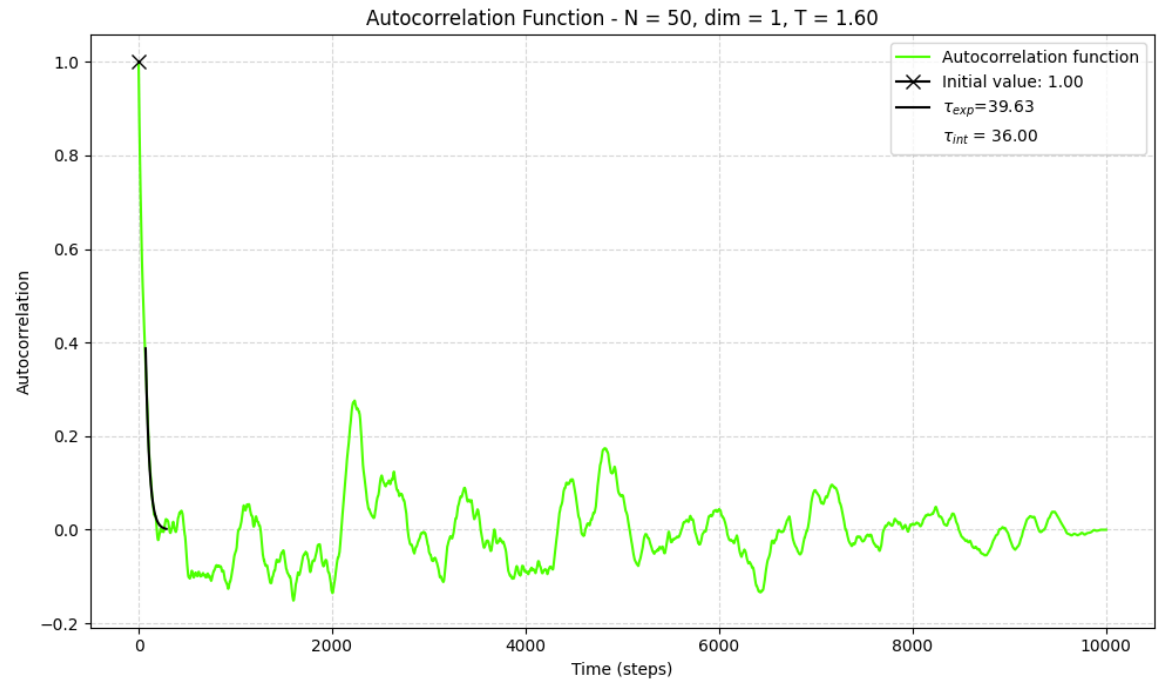


Figure 2.4: Fitting to find τ_{exp} .

[TODO: discussion of the data blocking method]

Chapter 3

Results

3.1 1D Ising model

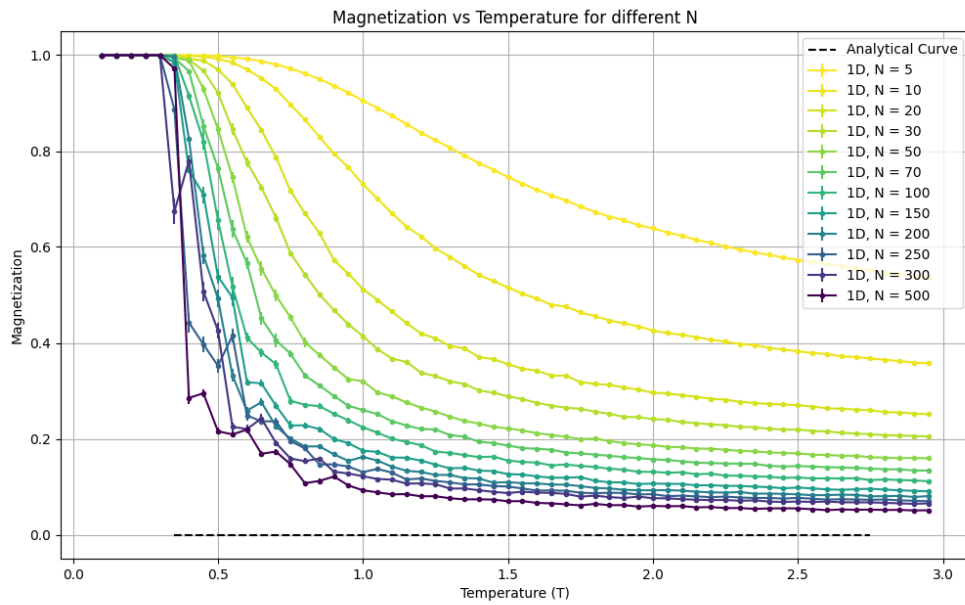


Figure 3.1: Convergence of magnetization for the 1D Ising model.

3.2 2D Ising model

3.3 3D Ising model

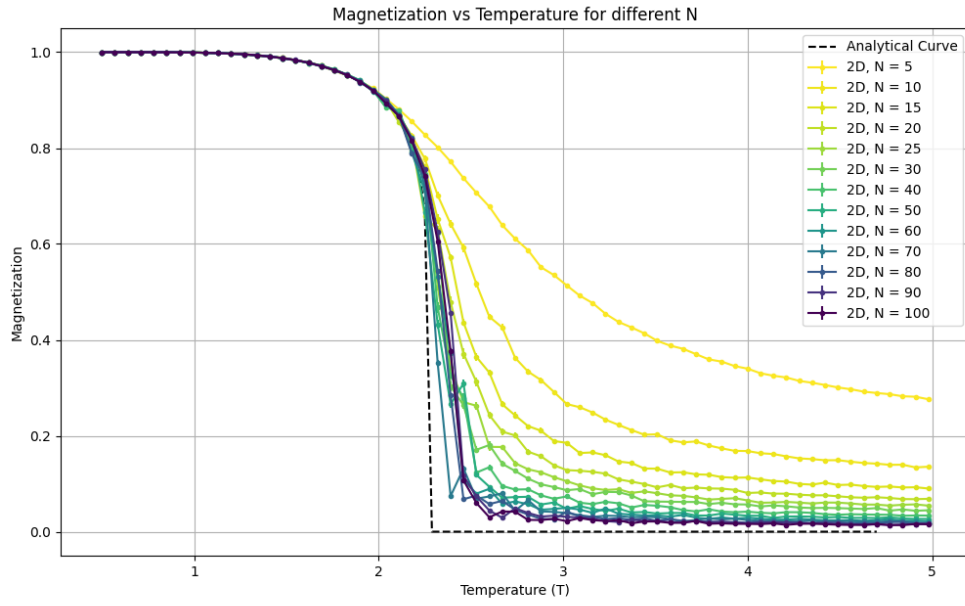


Figure 3.2: Convergence of magnetization for the 2D Ising model.

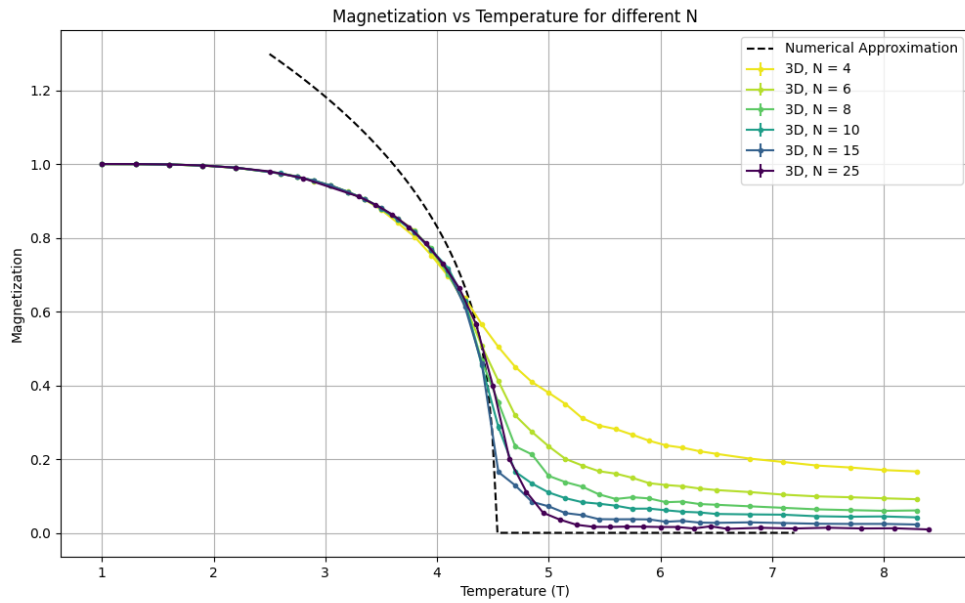


Figure 3.3: Convergence of magnetization for the 3D Ising model.

Appendix A

Appendix

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Bibliography

[1] Author, Title, Journal/Book, Year.